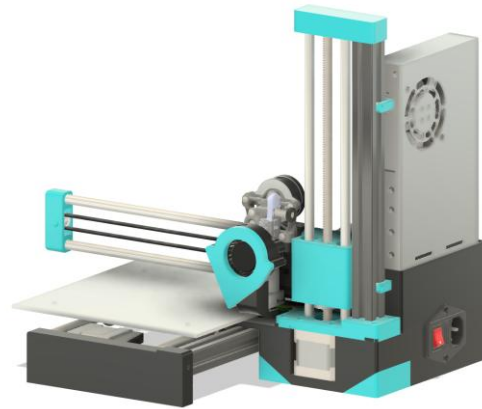
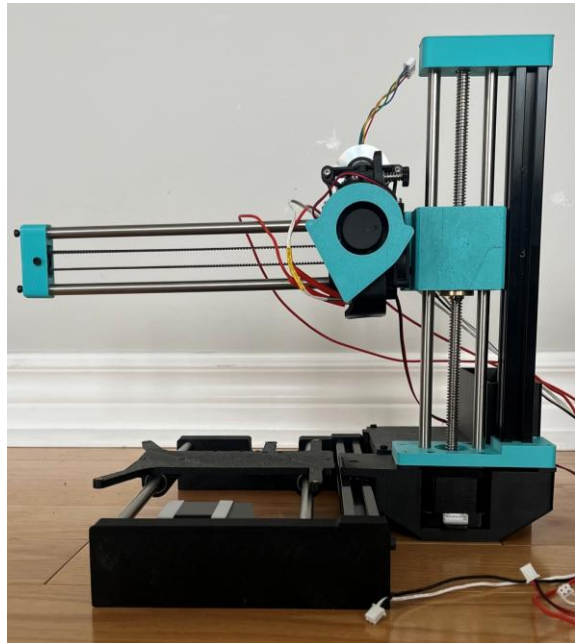
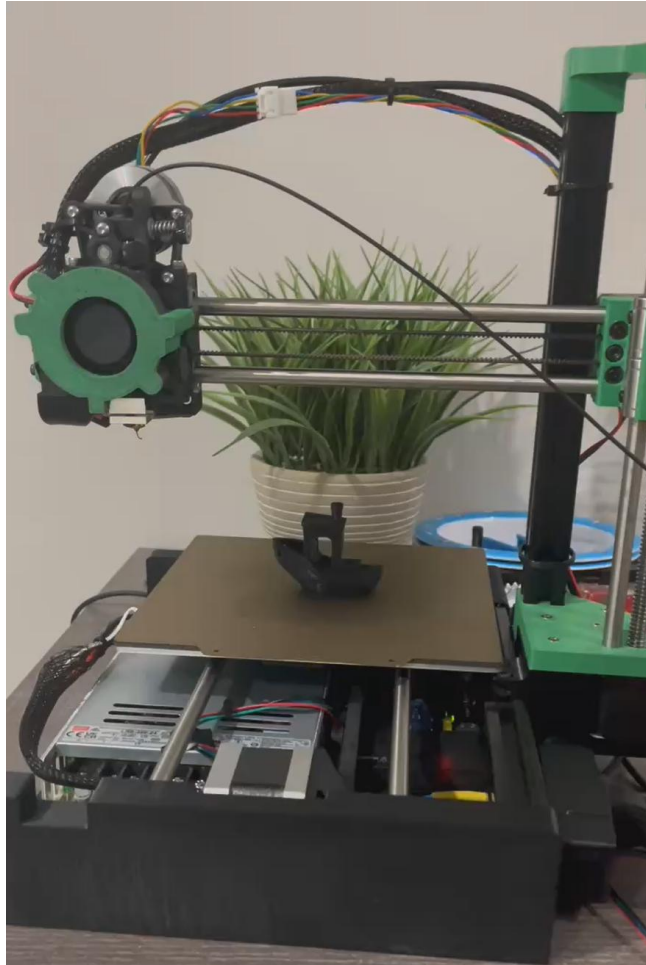


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Cantilever 3D Printer[↗]

February 2025 – August 2025

Task

- Design and build a reliable cantilever-style 3D printer with a 180x180x180mm build volume.
- Prioritize DFMA and a total cost less than \$300 USD.
- Incorporate eddy-based bed probing and design the system to support future input shaping.

Process

- Designed printer architecture using Fusion360.
- Created precise linear actuation systems using linear rods, belts, and leadscrews.
- Packaged toolhead with part-cooling fan and direct drive extruder, allowing for reliable and high-quality prints.

Result

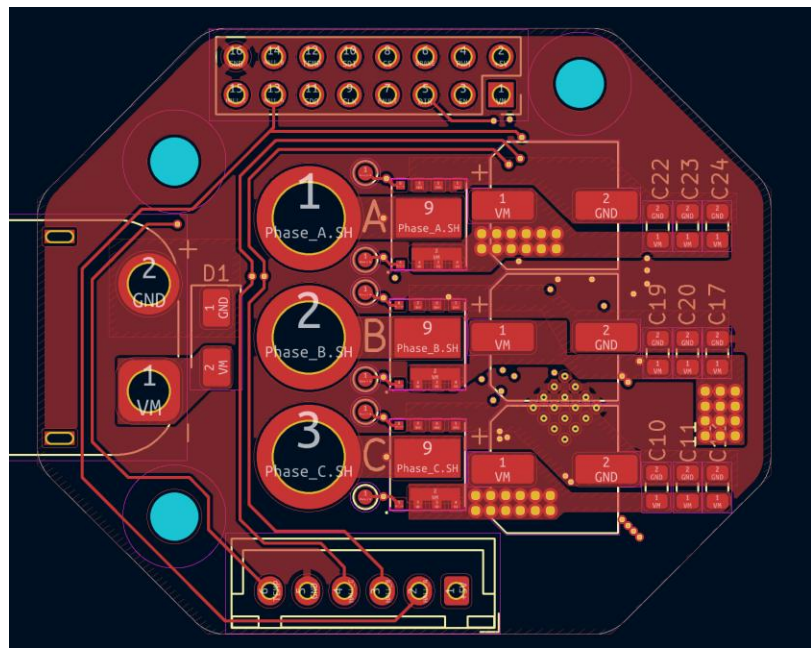
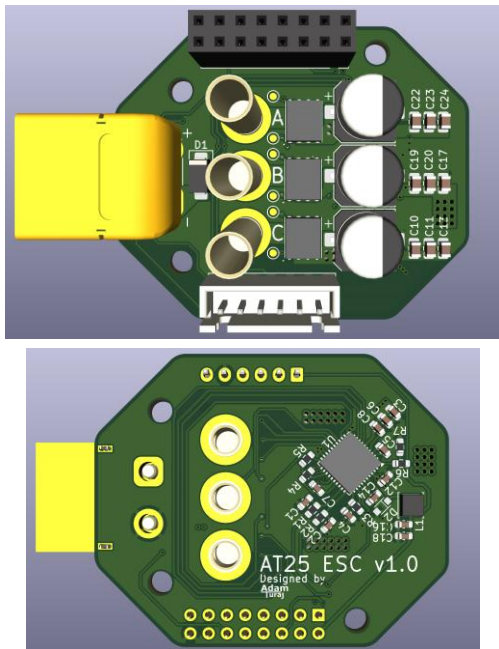
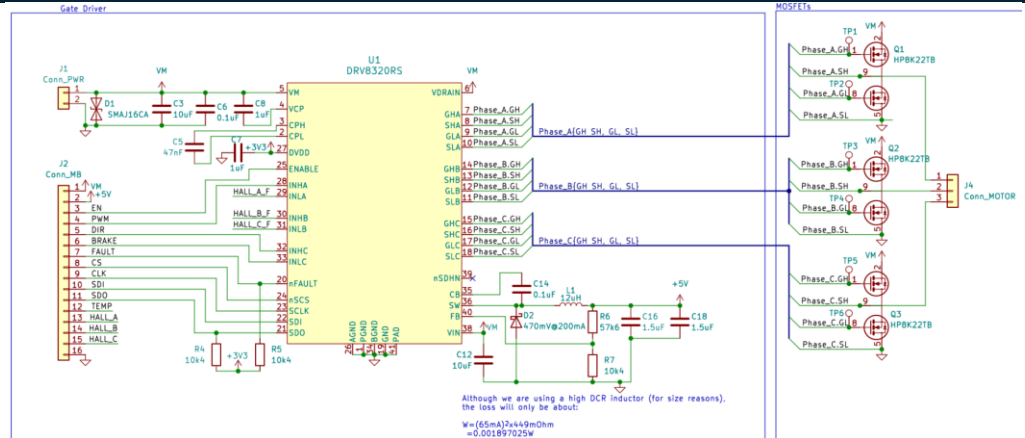
- Successfully printed a Benchy with comparable quality to off-the-shelf printers in 1 hour and 13 minutes.
- Achieved an easy assembly experience with proper tool clearance.
- Started building a V2 redesign, which showed promising improvements, but left it unfinished after university started.

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Gate Driver /
MOSFET
portion of the
schematic



Sensored BLDC ESC ↗

August 2025 – September 2025

Task

- Design a motor driver for a 2S sensored BLDC motor. Ensure the board can continuously supply 25A, with peak current capability up to 50A.
- Use the DRV8323 1x PWM control mode to simplify firmware requirements and avoid the need for complex field-oriented control algorithms.

Process

- Designed the schematic and 4-layer PCB layout in KiCad.
- Routed high-current motor power on a dedicated power layer with wide traces.
- Separated power and signal paths to reduce EMI and improve reliability.
- Added low-pass filters to the Hall-effect sensor inputs for cleaner feedback signals.

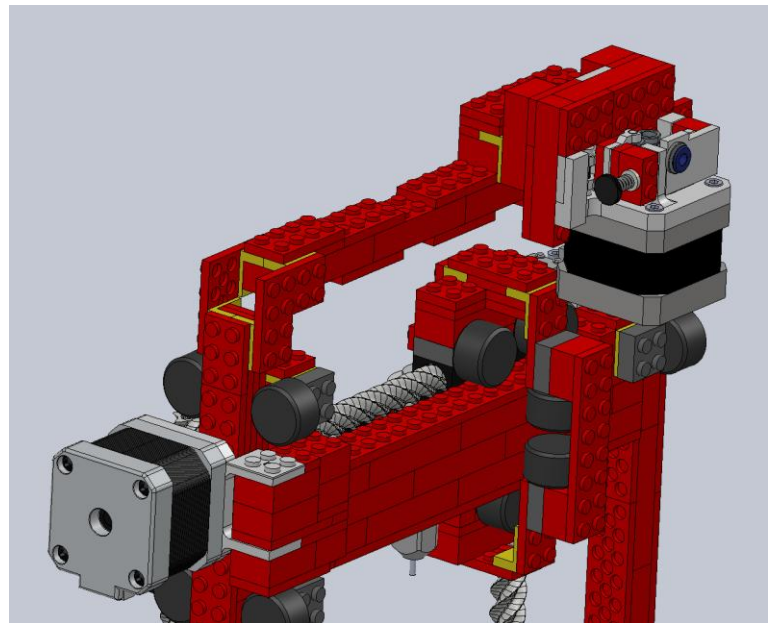
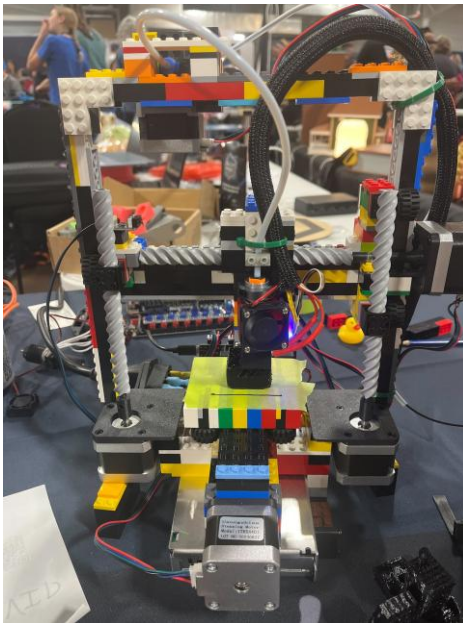
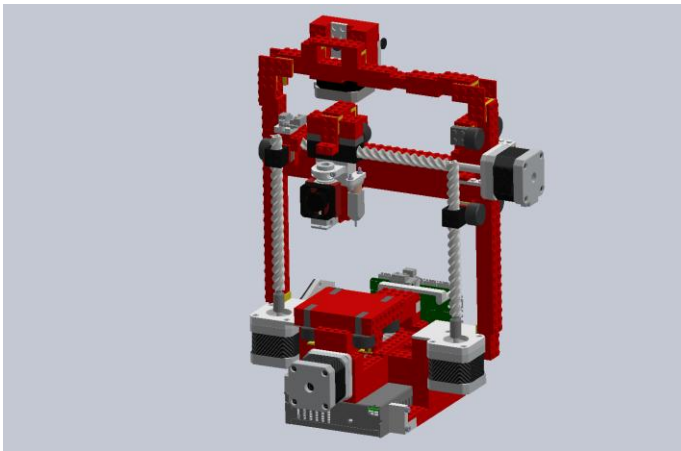
Result

- Routed all power, sensing, and communication lines within a compact 49 × 39mm PCB.
- Organized the layout for easier assembly, debugging, and understanding.
- Completed a fabrication-ready PCB design that demonstrates compact high-current ESC layout practices.

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LEGO 3D Printer[↗]

February 2026 – April 2026

Task

- Design a functional 3D printer built primarily from classic LEGO bricks, capable of printing a full-sized Benchy boat.
- Create a printer that is both mechanically sound and visually appealing.

Process

- Modeled the full mechanical system in SolidWorks.
- Used a Bowden-style extrusion system to reduce moving mass, making reliable printing more feasible.
- Designed custom 3D-printed mounts for the steppers and electronics.

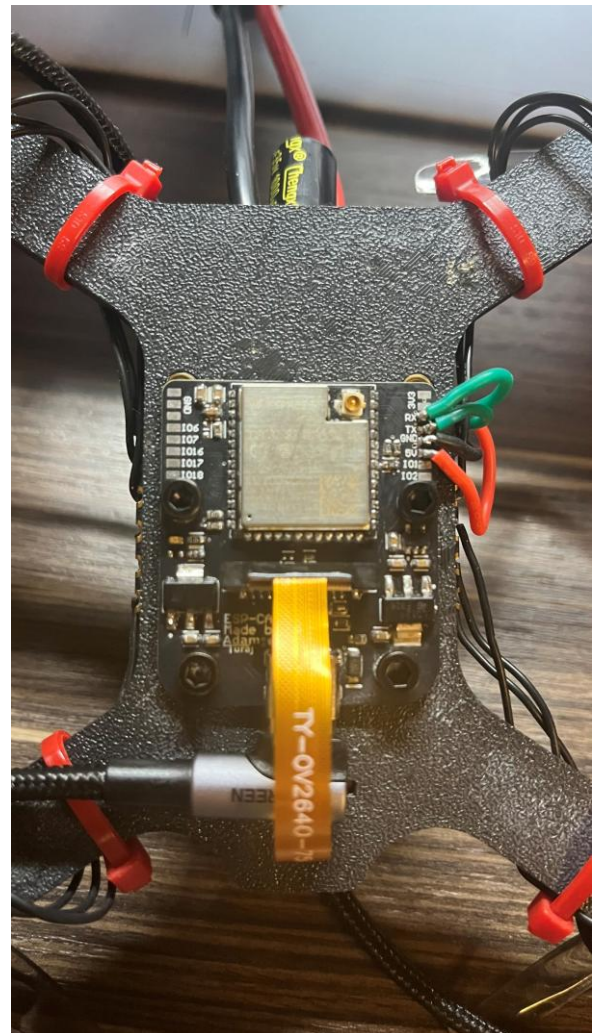
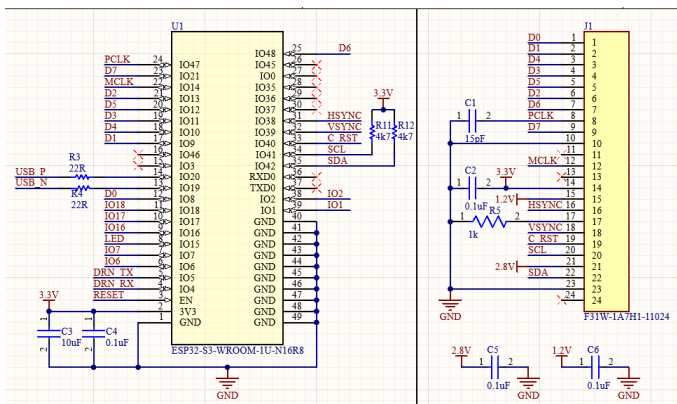
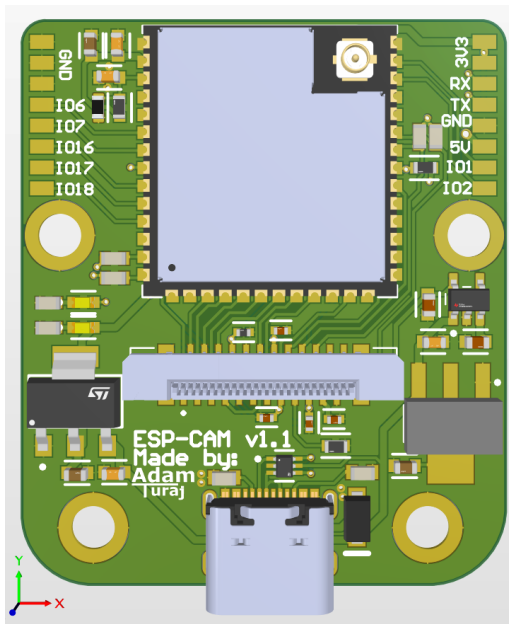
Result

- Successfully printed a full sized Benchy boat on the first attempt.
- Quality exceeded expectations with defects mainly caused by insufficient part cooling and slight Z-axis instability.
- Optimized design to use 92% LEGO by part count.

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Autonomous Drone[↗]

March 2026 – Present

Task

- Design a compact PCB integrating an ESP32 wireless module with a camera interface.
- Break out available GPIO pins for future expansion and debugging.

Process

- Traced a simple 2-layer PCB in Altium to fit flat under the drone.
- Prioritized short, continuous camera data traces for reliable signal routing.
- Added voltage regulation to properly supply the camera module.
- Selected decoupling capacitors to improve camera clock and signal stability.

Result

- Successful flashing and wireless testing of ESP32 and wireless capabilities.
- Achieved communication with flight control unit over UART.
- Captured and streamed camera frames reliably over UDP.

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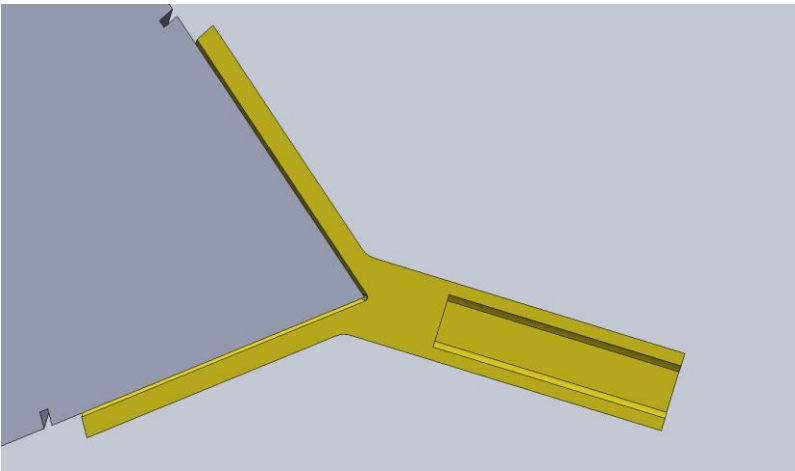
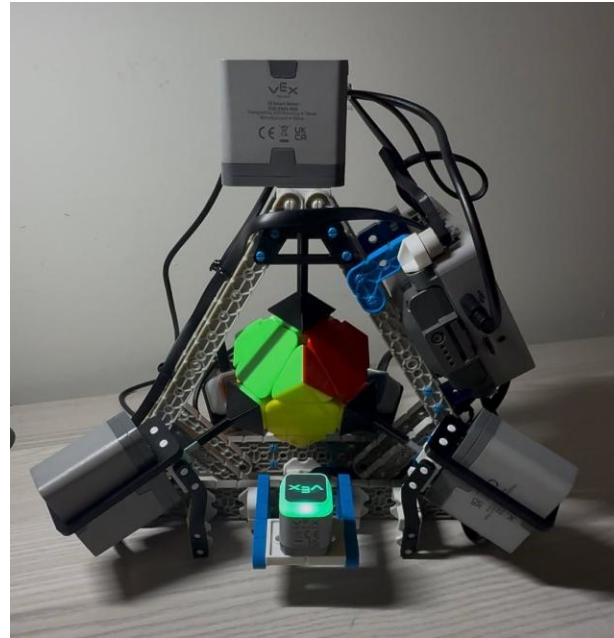
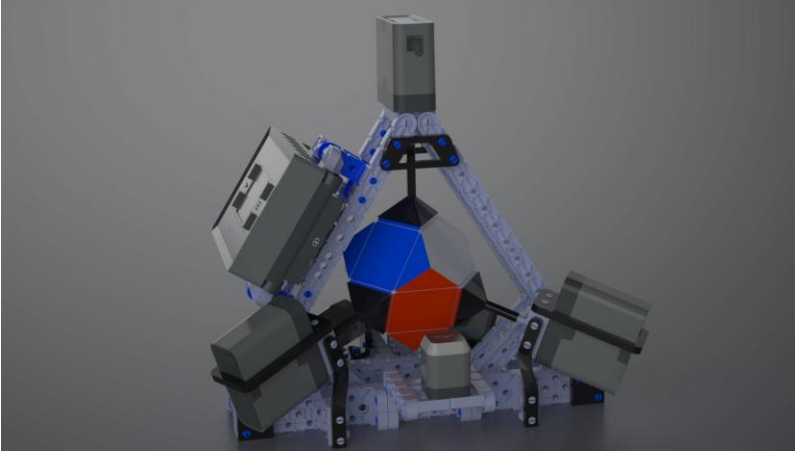
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Skewb Solving Robot[↗]

October 2025 – December 2025

Task

- Design and build a robot using mostly VEX parts and electronics.
- Create a robot capable of solving a Skewb in under 1 minute with an 80% solve success rate.
- Write a technical report outlining project scope and design process.
- Collaborate with a four-person team to keep the project on schedule and meet deadlines.

Process

- Used SolidWorks to design the structure and custom 3D-printed parts.
- Wrote and helped tune a PID algorithm to improve the number of successful moves and increase solve speed.
- Refined the design to improve reliability by adjusting gripper geometry and adding additional bracing to the motors.

Result

- The Skewbot met and exceeded project goals, achieving solve times as fast as 13 seconds and a solve success rate of approximately 80%.
- The robot was presented to TAs, professors, and peers, who provided positive feedback.